Senior Data Engineer

Thank you for taking the time to participate in Jua.ai's Geospatial Data Engineering assessment.
This challenge is intended to give us deeper insight into your skill level and approach to problem-solving.

Context

At Jua, we work extensively with multi-dimensional weather data. Much of this data is stored as large multidimensional arrays in formats like grib2, NetCDF or HDF5. While these formats are efficient for storing vast amounts of meteorological data, they are not always straightforward for performing ad-hoc data analysis. We often transform these raw files into a standardized format to be available for further processing steps.

Task Overview

You will build a basic ETL pipeline to:

1. **Extract and transform** the total precipitation data for the **year 2022** from the ERA5 dataset (link provided below).
2. **Add a hierarchical geospatial index** (e.g., using H3 or another open-source indexing method of your choice) to the extracted data.
3. Load the transformed data into **Apache Parquet.**
4. **Demonstrate sample queries** that show how to filter this data by:
    - Time range (e.g., specific timestamps or intervals)
    - Geospatial location (using the chosen hierarchical index)

This assignment is intentionally scoped to be completed in approximately 4 hours. We do not expect a fully productionized system; rather, we want to see your reasoning, approach, and coding style.

Source Data

- **Data Location:**
  Source data for the task can be downloaded from the ARCO data set, which is hosted here:
  `gs://gcp-public-data-arco-era5/raw/date-variable-single_level/`
- **Variable of Interest:** Total precipitation data for the year 2022.

You can use any open-source tools and frameworks. We can, however, ask you to explain your choice.

Requirements and submission guidelines

- The solution can be implemented in any modern programming language.
- Please consider code quality and readability.
- Provide a README with instructions on how to set up, run, and test your app.
- Submission can be a ZIP file or repository hosted in a public location (e.g., Github).